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|----------------------|--------------|
| <b>Date</b>          | 23 July 2026 |
| <b>Project Name</b>  | SB-STOCKS    |
| <b>Maximum Marks</b> | 5            |

## Solution Requirements

### Functional Requirements:

1. User Registration & Login with JWT authentication
2. Stock listing with real-time market data
3. Buy/Sell stocks with virtual currency
4. Portfolio dashboard showing holdings & P&L
5. Watchlist management for tracking favorite stocks
6. Admin panel for stock management
7. Transaction history & analytics
8. Search & filter stocks

### Non-Functional Requirements:

1. Response time < 2 seconds for all API calls
2. Secure password hashing with bcrypt
3. Mobile-responsive design
4. Support for 100+ concurrent users
5. Data consistency in transactions

### Software Requirements

| S.No | Requirement      | Technology          | Version   |
|------|------------------|---------------------|-----------|
| 1    | Operating System | Windows/macOS/Linux | Win 10/11 |
| 2    | Runtime          | Node.js             | v16+      |
| 3    | Package Manager  | npm                 | v8+       |
| 4    | Frontend         | React.js            | v18+      |
| 5    | Backend          | Express.js          | v4+       |
| 6    | Database         | MongoDB             | v6+       |
| 7    | Browser          | Chrome/Firefox      | Latest    |
| 8    | Code Editor      | VS Code             | Latest    |
| 9    | API Testing      | Postman             | Latest    |

|    |                 |              |        |
|----|-----------------|--------------|--------|
| 10 | Version Control | Git & GitHub | Latest |
|----|-----------------|--------------|--------|

### Hardware Requirements

| S.No | Component | Minimum                        | Recommended            |
|------|-----------|--------------------------------|------------------------|
| 1    | Processor | Intel i5 8th Gen / AMD Ryzen 5 | Intel i7 / AMD Ryzen 7 |
| 2    | RAM       | 8 GB                           | 16 GB                  |
| 3    | Storage   | 1 GB free                      | 5 GB free              |
| 4    | Display   | 1366x768                       | 1920x1080              |